

Chap. 5

Applications of Newton's Laws

Limitations of Newton's laws

- In the macro-scale

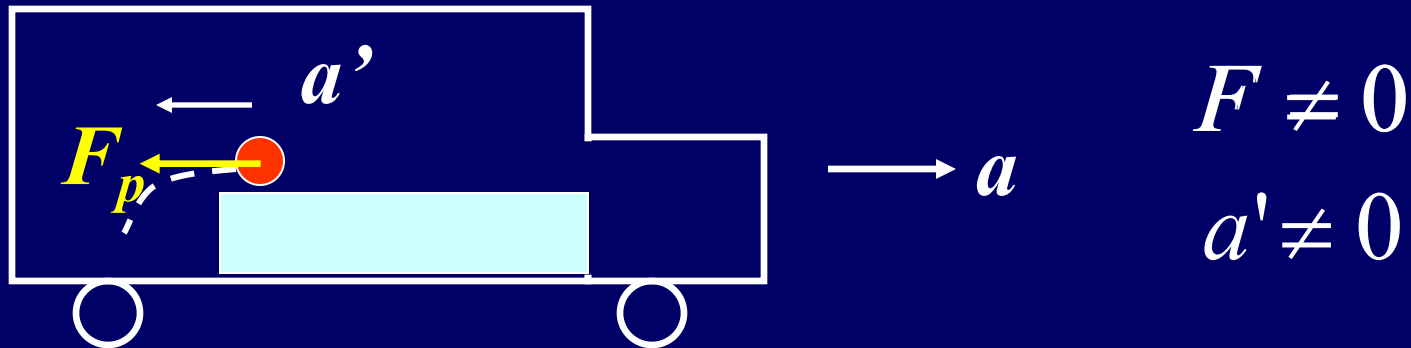
- Inertia frame

its velocity is constant relative to one another

- If a frame S' moves with acceleration, it is not an inertial frames.

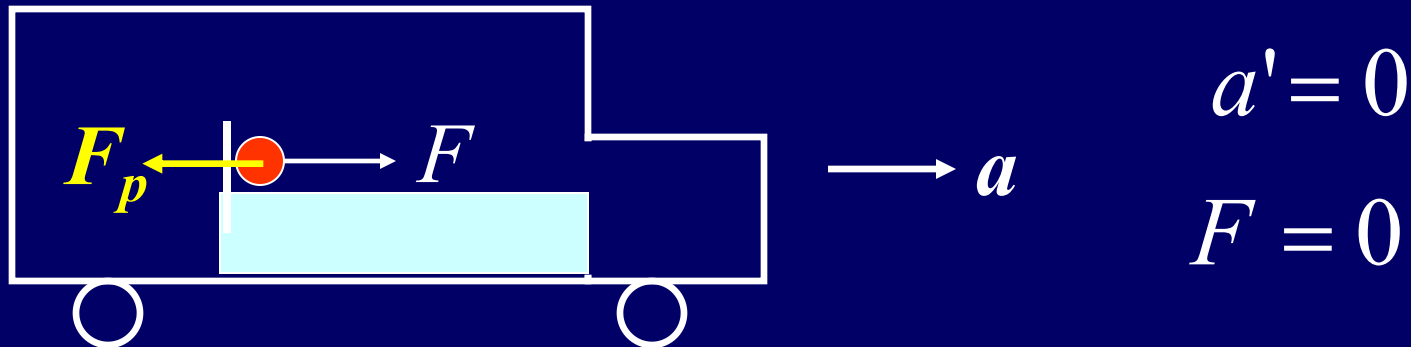
Noninertial frames and pseudoforces

Noninertial frames and pseudoforces



$$F \neq 0$$

$$a' \neq 0$$

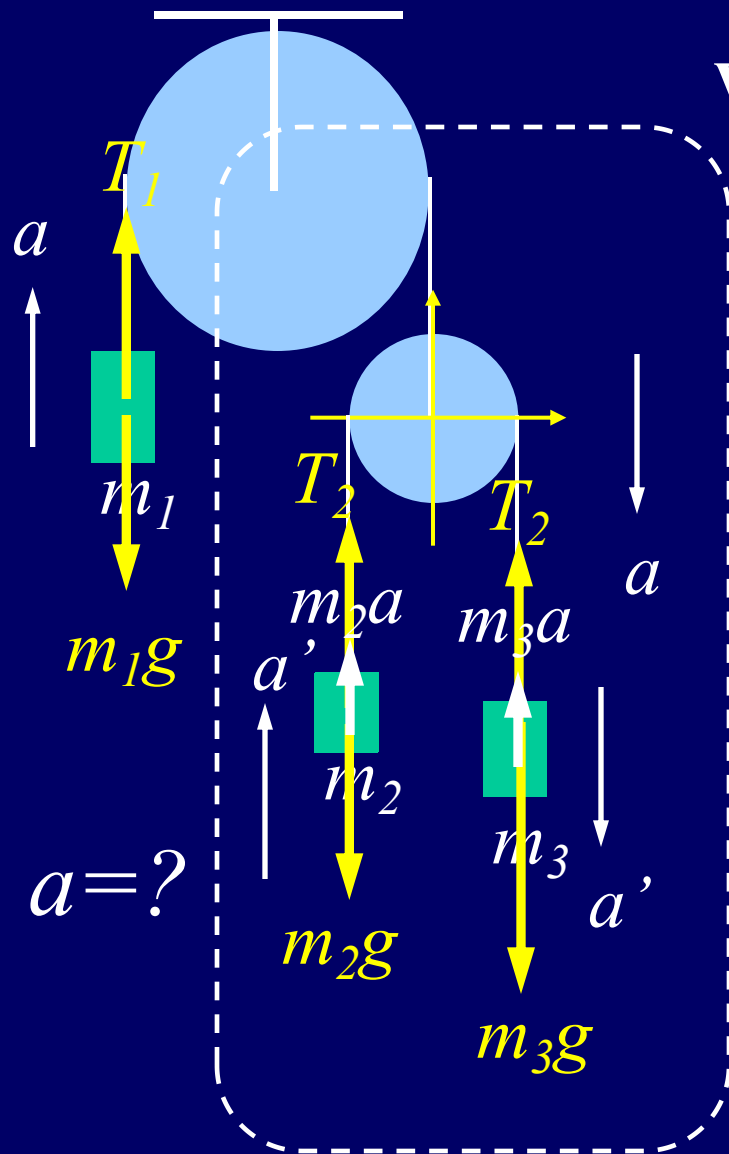


$$a' = 0$$

$$F = 0$$

Pseudoforce $F_p = -ma$ **inertial force**

$$\Sigma \vec{F} = \Sigma \vec{F}_0 + \vec{F}_p = m\vec{a}'$$



We assume $m_1 < m_2 + m_3$ and $m_2 < m_3$

Noninertial Frame $T_1 - m_1 g = m_1 a$

Pseudoforce, $(-ma)$ $T_1 = 2 T_2$

$$T_2 + m_2 a - m_2 g = m_2 a'$$

$$m_3 g - m_3 a - T_2 = m_3 a'$$

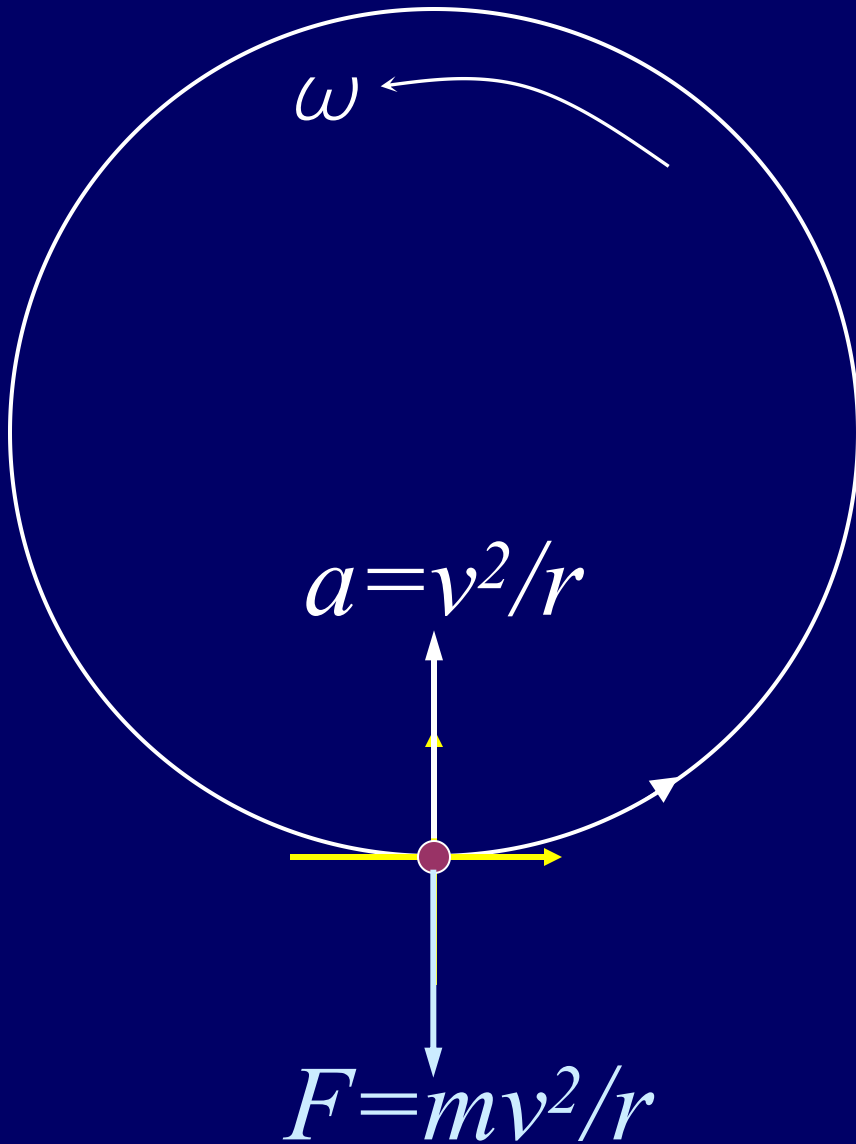
$$a = \frac{4m_2 m_3 - m_1 m_2 - m_1 m_3}{4m_2 m_3 + m_1 m_2 + m_1 m_3} g$$

$$a' = \frac{2m_1(m_3 - m_2)}{4m_2 m_3 + m_1 m_2 + m_1 m_3} g$$

$$a_2 = a - a' \quad a_3 = a + a'$$

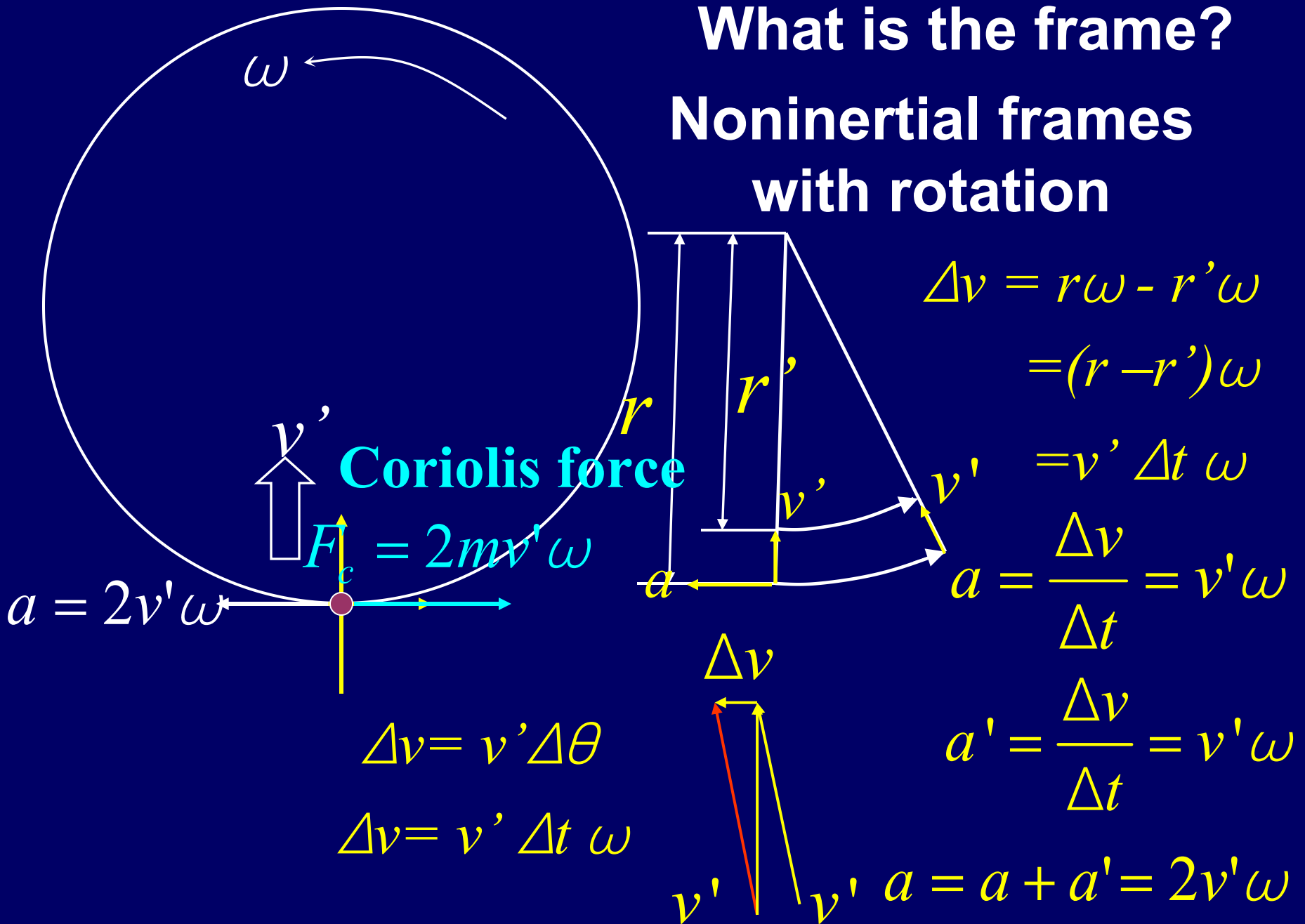
What is the frame?

Noninertial frames with rotation



What is the frame?

Noninertial frames with rotation



What is the frame?

Noninertial frames with rotation

$$\vec{v} = -v' \hat{u}_r + r\omega \hat{u}_\phi$$

$$\frac{d\vec{v}}{dt} = -v' \frac{d\hat{u}_r}{dt} + \frac{d(r\omega)}{dt} \hat{u}_\phi + r\omega \frac{d\hat{u}_\phi}{dt}$$

$$\frac{d\vec{v}}{dt} = -v' \frac{d\phi}{dt} \hat{u}_\phi + \omega \frac{dr}{dt} \hat{u}_\phi - r\omega \frac{d\phi}{dt} \hat{u}_r$$

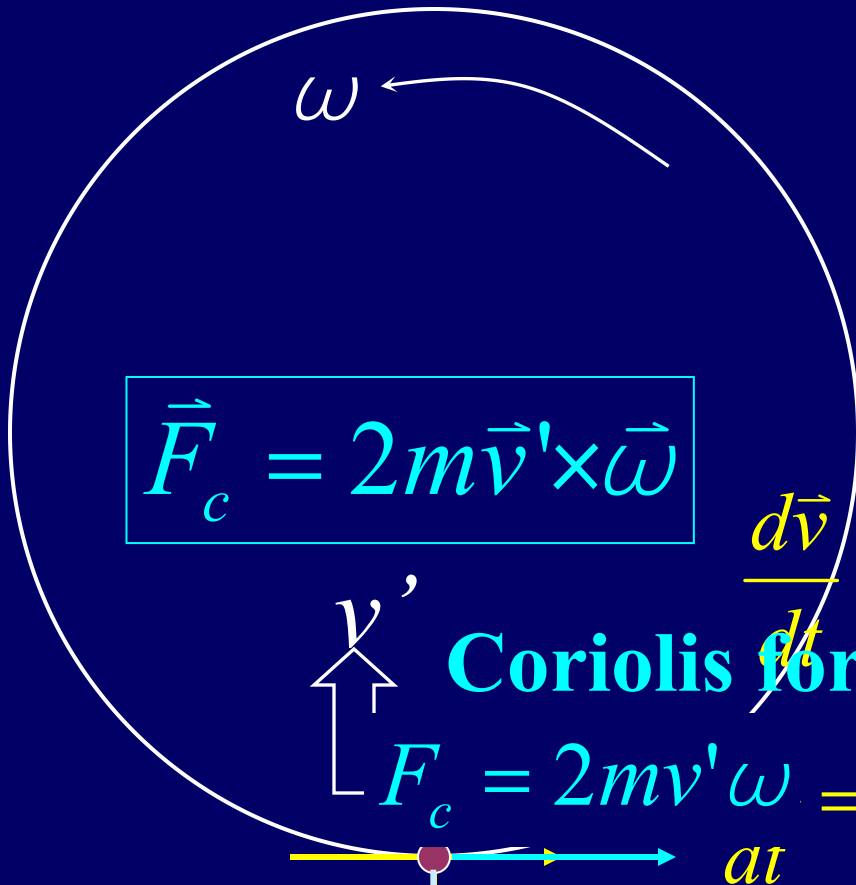
$$\frac{d\vec{v}}{dt} = -2v' \omega \hat{u}_\phi - \frac{v'^2}{r} \hat{u}_r$$

$$\vec{F}_c = 2m\vec{v}' \times \vec{\omega}$$

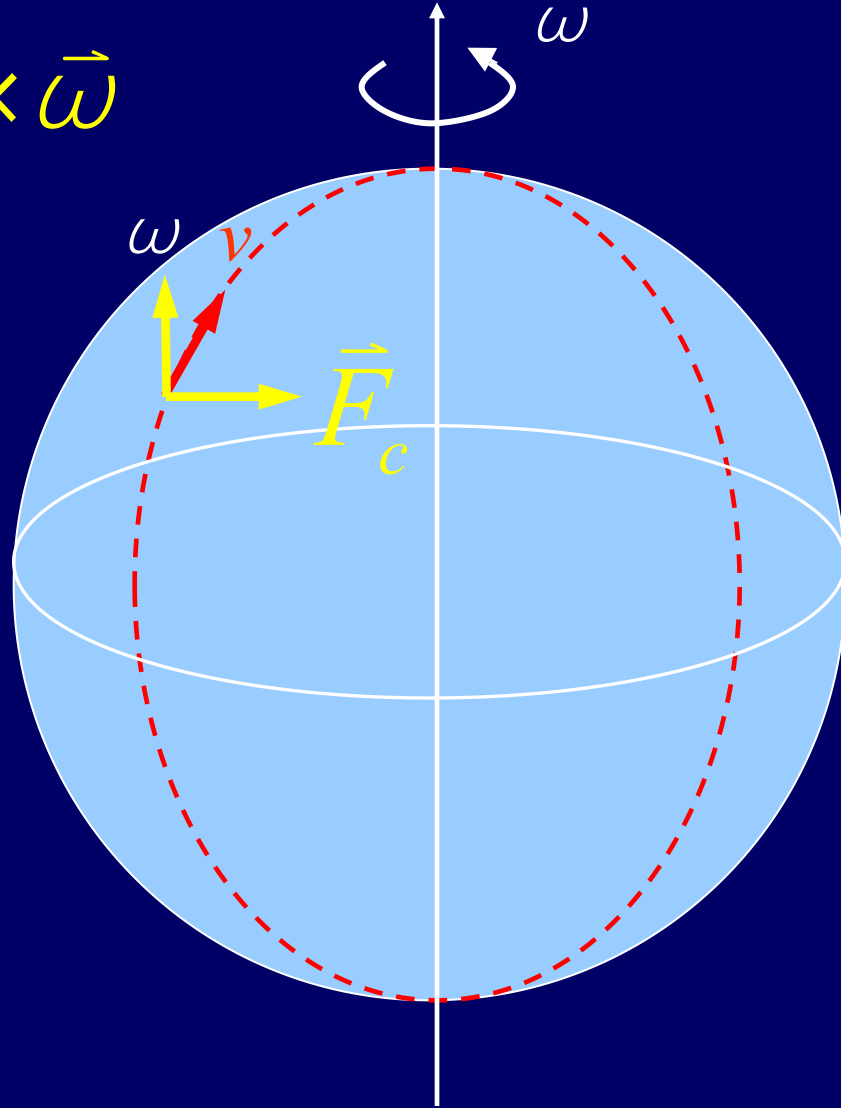
Coriolis force

$$F_c = 2mv' \omega$$

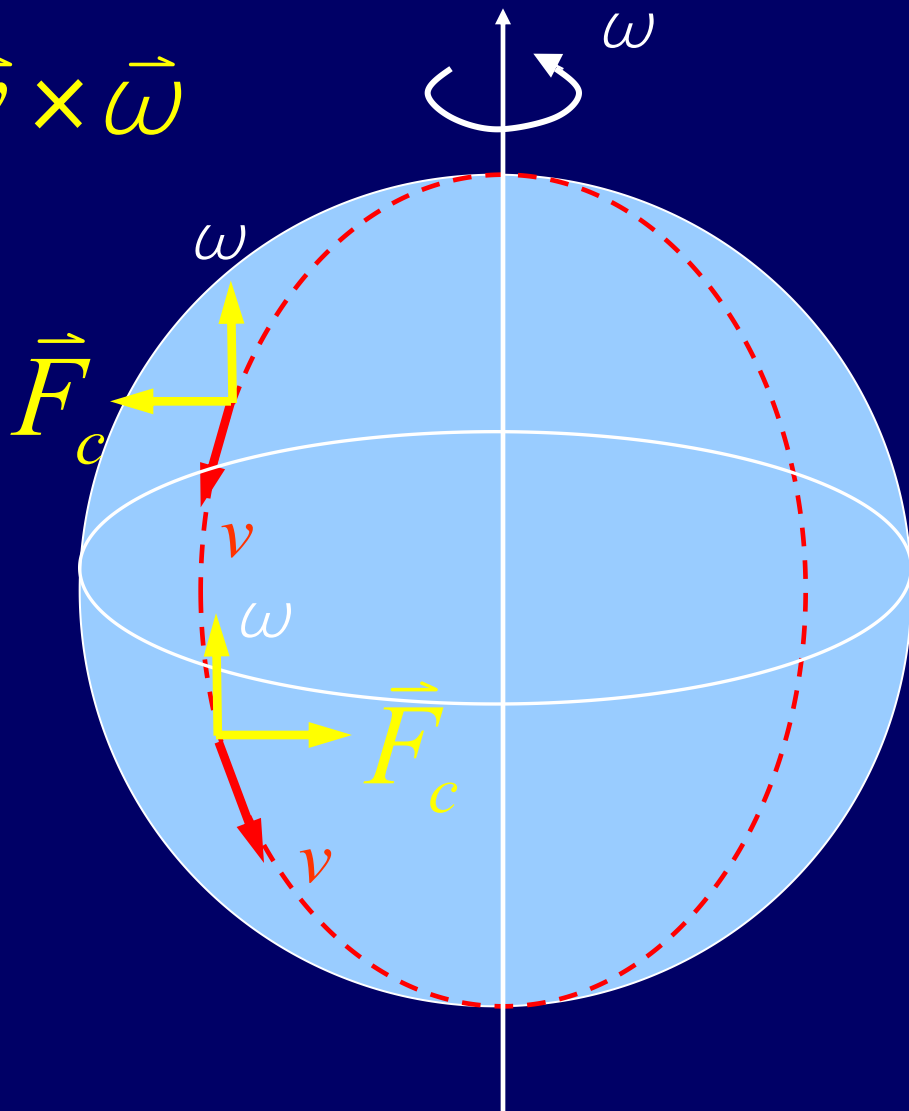
$$F = mv^2/r$$



$$\vec{F}_c = 2m\vec{v} \times \vec{\omega}$$



$$\vec{F}_c = 2m\vec{v} \times \vec{\omega}$$



Chap. 5

Exercises

11 (P110), 18 (P111), 39 (P113)

Problems

17, 18 (P116)